



ĐẠI HỌC BÁCH KHOA HÀ NỘI
VIỆN CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG

Mashup-based Linked Data Integration

Agenda

1. Introduction
2. Linked Widgets Framework
3. Mashup Models
4. Conclusion

1. Introduction

Motivation

CSV(lat, long, name, etc.)

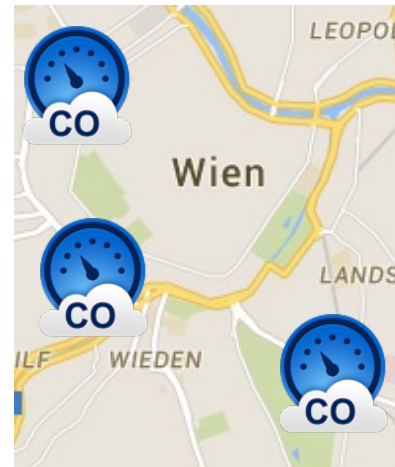
XML(lat, long, Image URL)

JSON(lat, long, CO, NO₂, etc.)

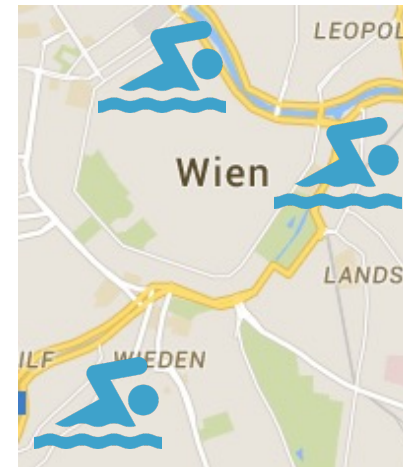
CSV(lat, long, name, etc.)



Flickr/Google
Image Search



Air quality data



51 public swimming
pools in Vienna

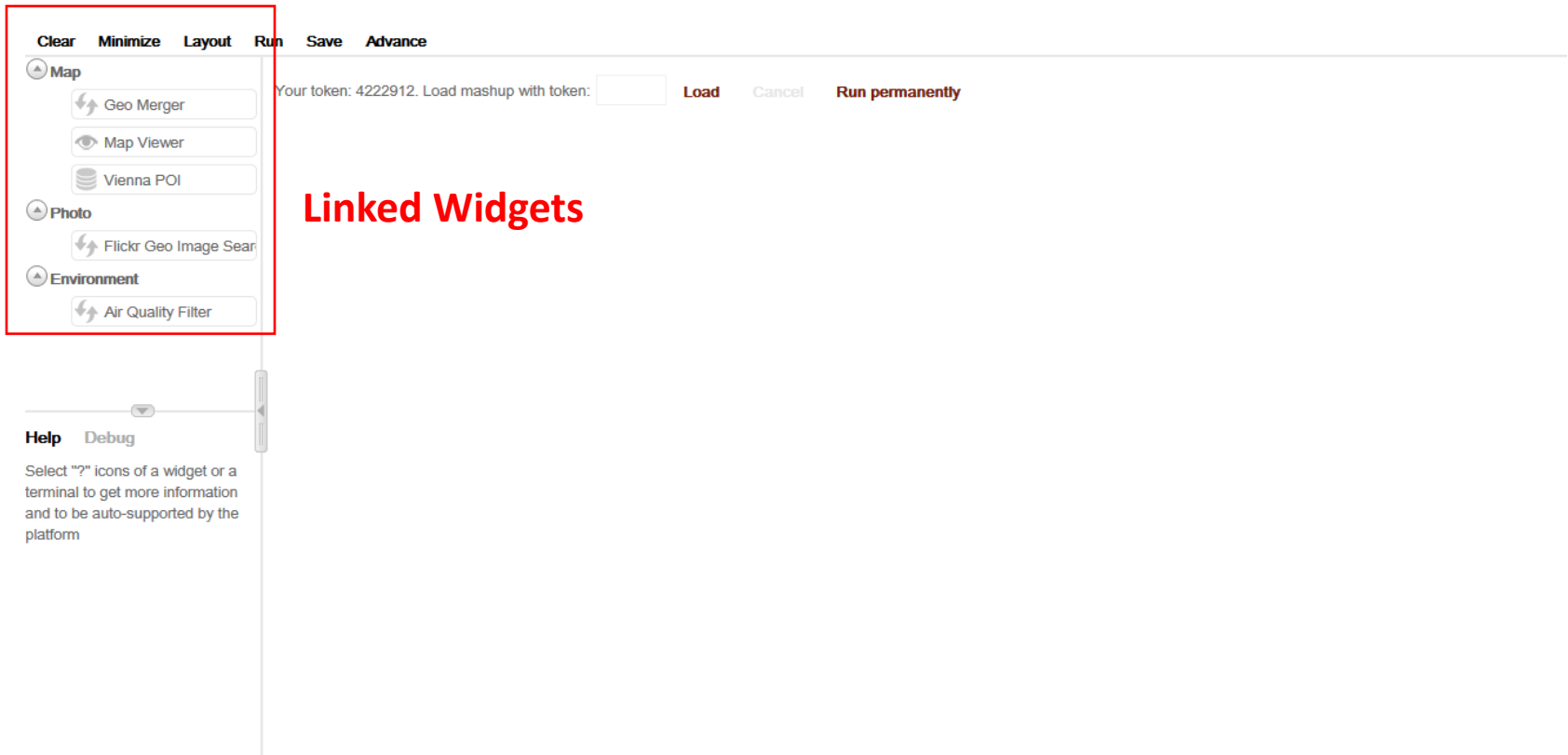


?

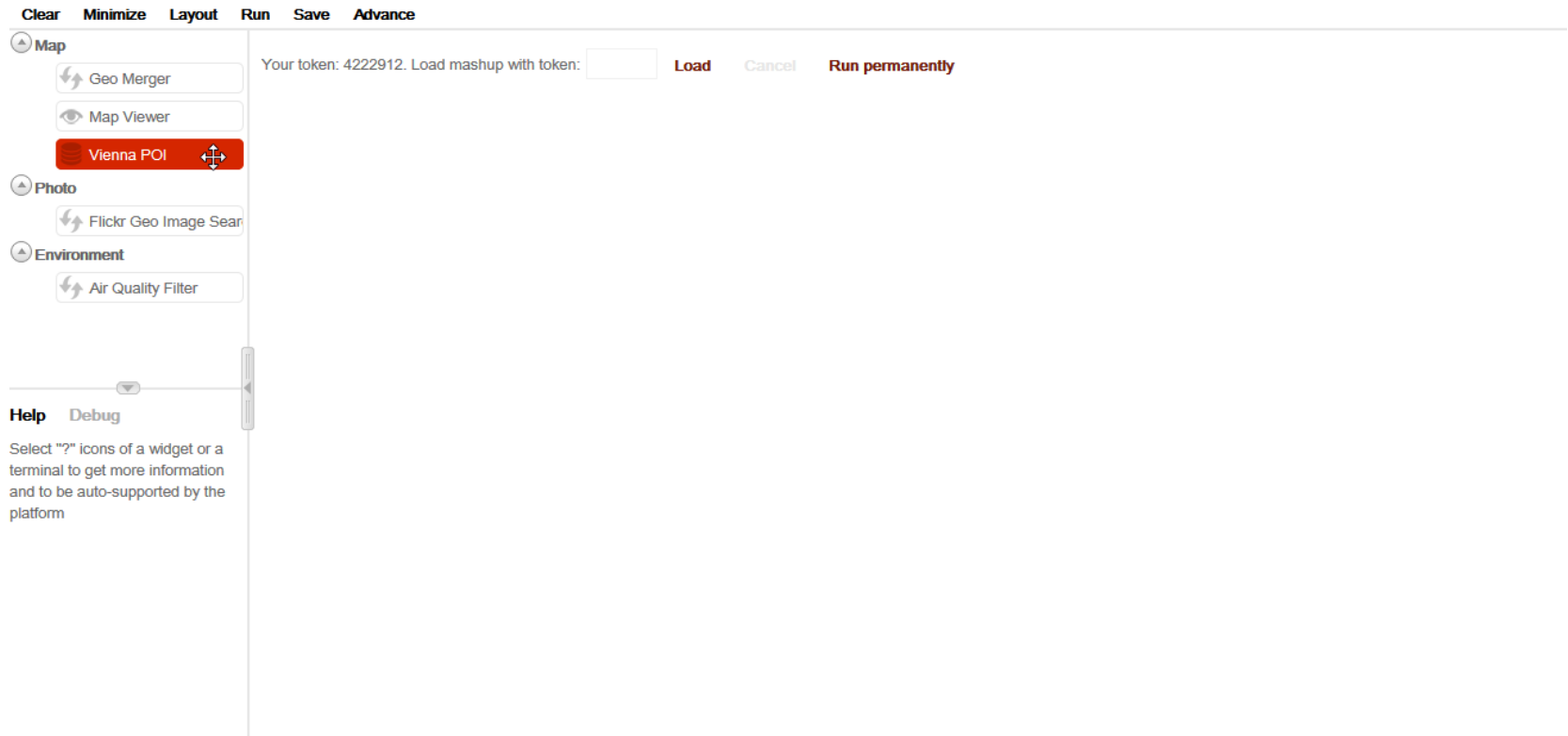
Motivation



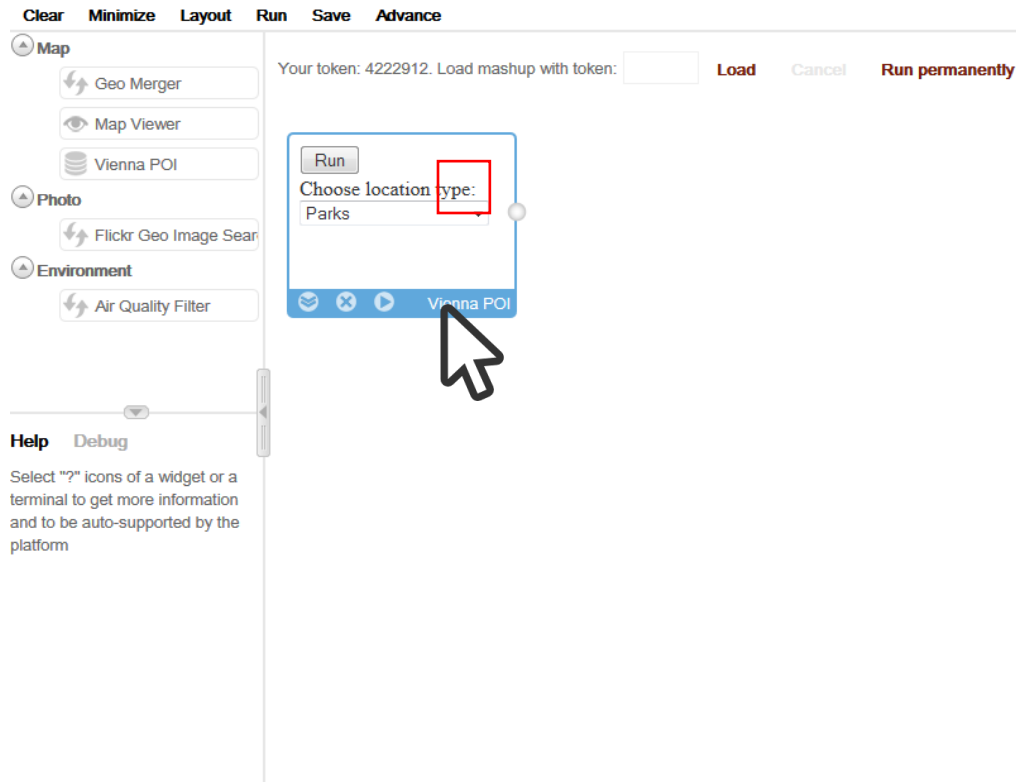
Modular approach for data integration



Modular approach for data integration



Modular approach for data integration



Modular approach for data integration

The screenshot displays a web-based mashup development environment. At the top, a menu bar includes 'Clear', 'Minimize', 'Layout', 'Run', 'Save', and 'Advance'. On the left sidebar, there are three main categories: 'Map' (containing 'Geo Merger', 'Map Viewer', and 'Vienna POI'), 'Photo' (containing 'Flickr Geo Image Search'), and 'Environment' (containing 'Air Quality Filter'). Below these is a 'Help' section with a 'Debug' button and a text box explaining that '?' icons provide more information and auto-support. The main workspace shows a 'Run' button, a 'Choose location type:' dropdown menu with 'Parks' selected, and a 'Vienna POI' button. Above the workspace, a text field displays 'Your token: 4222912. Load mashup with token:' followed by 'Load', 'Cancel', and 'Run permanently' buttons.

Modular approach for data integration

Clear Minimize Layout Run Save Advance

Map

- Geo Merger
- Map Viewer
- Vienna POI

Photo

- Flickr Geo Image Search

Environment

- Air Quality Filter

Your token: 4222912. Load mashup with token: Load Cancel Run permanently

Run

Choose location type:

Parks

Run

Minimum Level:

Very good

Maximum Level:

None

Vienna POI

Air Quality Filter

Help Debug

Select "?" icons of a widget or a terminal to get more information and to be auto-supported by the platform

Modular approach for data integration

The screenshot displays a web-based data integration platform interface. At the top, there is a navigation bar with buttons: Clear, Minimize, Layout, Run, Save, and Advance. Below this, a sidebar on the left contains a list of widgets categorized under 'Map', 'Photo', and 'Environment'. The 'Map' category includes 'Geo Merger', 'Map Viewer', and 'Vienna POI'. The 'Photo' category includes 'Flickr Geo Image Search'. The 'Environment' category includes 'Air Quality Filter'. A 'Help' section at the bottom of the sidebar provides instructions: 'Select "?" icons of a widget or a terminal to get more information and to be auto-supported by the platform'. The main workspace shows a workflow with two widgets: 'Vienna POI' and 'Air Quality Filter'. The 'Vienna POI' widget has a 'Run' button and a 'Choose location type:' dropdown menu with 'Parks' selected. The 'Air Quality Filter' widget has a 'Run' button, a 'Minimum Level:' dropdown menu with 'Very good' selected, and a 'Maximum Level:' dropdown menu with 'None' selected. A mouse cursor is pointing at the 'Parks' option in the 'Choose location type:' dropdown. Above the widgets, there is a text input field for a token, currently containing '4222912', and buttons for 'Load', 'Cancel', and 'Run permanently'.

Clear Minimize Layout Run Save Advance

Map

- Geo Merger
- Map Viewer
- Vienna POI

Photo

- Flickr Geo Image Search

Environment

- Air Quality Filter

Help Debug

Select "?" icons of a widget or a terminal to get more information and to be auto-supported by the platform

Your token: 4222912. Load mashup with token: Load Cancel Run permanently

Run

Choose location type:

Parks

Run

Minimum Level:

Very good

Maximum Level:

None

Vienna POI

Air Quality Filter

Modular approach for data integration

Clear Minimize Layout Run Save Advance

Map

- Geo Merger
- Map Viewer
- Vienna POI

Photo

- Flickr Geo Image Set

Environment

- Air Quality Filter

Your token: 4222912. Load mashup with token: Load Cancel Run permanently

Run

Choose location type:

Parks

Run

Minimum Level:

Very good

Maximum Level:

None

Vienna POI

Air Quality Filter

Help Debug

Select "?" icons of a widget or a terminal to get more information and to be auto-supported by the platform

Modular approach for data integration

Clear Minimize Layout Run Save Advance

Map

- Geo Merger
- Map Viewer
- Vienna POI

Photo

- Flickr Geo Image Search

Environment

- Air Quality Filter

Help Debug

Select "?" icons of a widget or a terminal to get more information and to be auto-supported by the platform

Your token: 4222912. Load mashup with token: Load Cancel Run permanently

Run

Choose location type:
Parks

Run

Minimum Level:
Very good

Maximum Level:
None

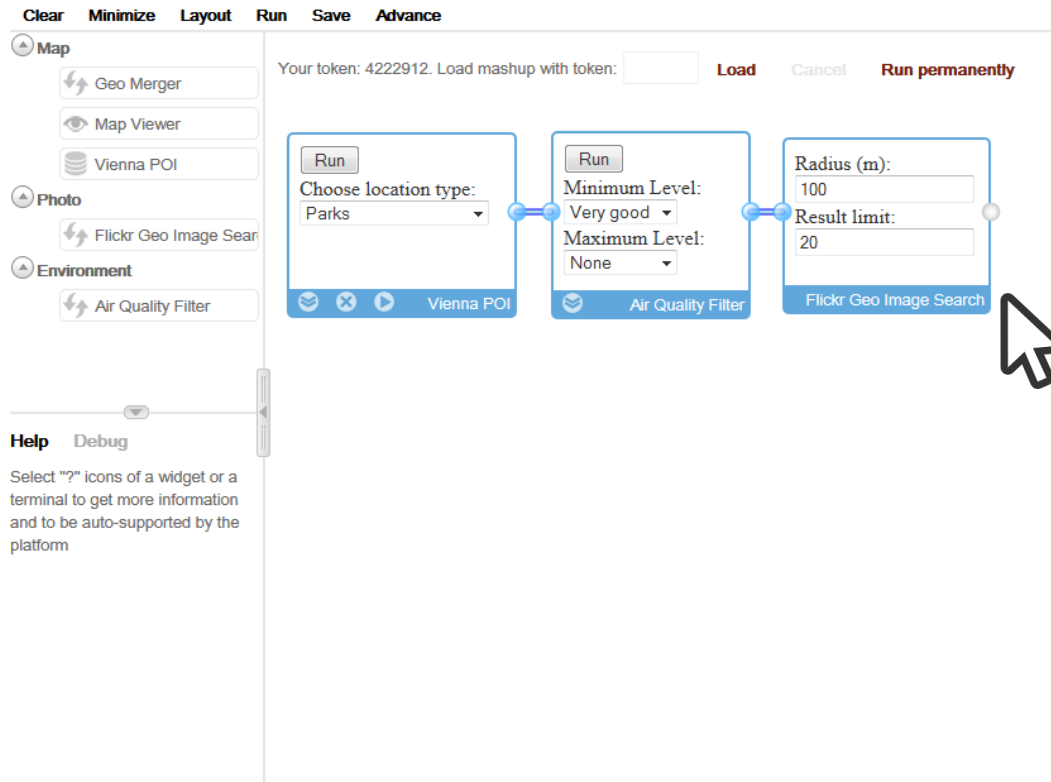
Radius (m):
100

Result limit:
20

Vienna POI

Air Quality Filter

Flickr Geo Image Search



Modular approach for data integration

Clear Minimize Layout Run Save Advance

Map

- Geo Merger
- Map Viewer
- Vienna POI**

Photo

- Flickr Geo Image Search

Environment

- Air Quality Filter

Your token: 4222912. Load mashup with token: **Load** **Cancel** **Run permanently**

Run Choose location type: Parks

Run Minimum Level: Very good Maximum Level: None

Radius (m): 100 Result limit: 20

Vienna POI Air Quality Filter Flickr Geo Image Search

Help Debug

Select "?" icons of a widget or a terminal to get more information and to be auto-supported by the platform

Modular approach for data integration

Clear Minimize Layout Run Save Advance

Map

- Geo Merger
- Map Viewer
- Vienna POI

Photo

- Flickr Geo Image Search

Environment

- Air Quality Filter

Your token: 4222912. Load mashup with token: Load Cancel Run permanently

Run

Choose location type:
Parks

Run

Minimum Level:
Very good

Maximum Level:
None

Radius (m):
100

Result limit:
20

Vienna POI

Air Quality Filter

Flickr Geo Image Search

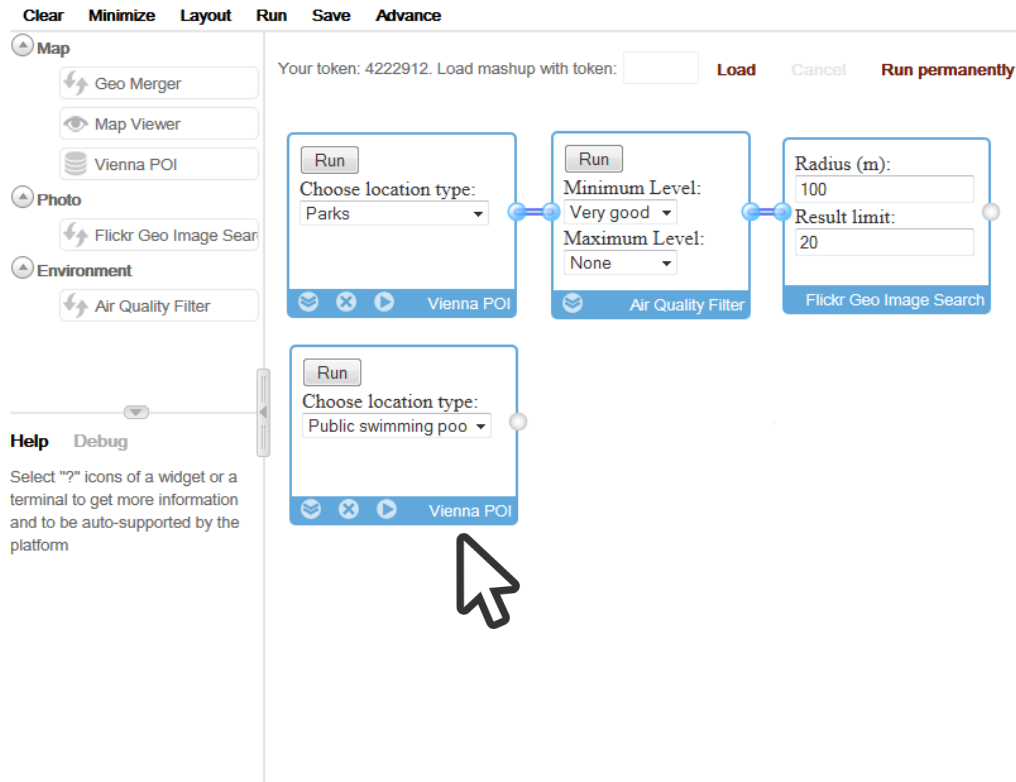
Run

Choose location type:
Public swimming pool

Vienna POI

Help Debug

Select "?" icons of a widget or a terminal to get more information and to be auto-supported by the platform



Modular approach for data integration

Clear Minimize Layout Run Save Advance

Map

Geo Merger

Map Viewer

Vienna POI

Photo

Flickr Geo Image Search

Environment

Air Quality Filter

Your token: 4222912. Load mashup with token: Load Cancel Run permanently

Run

Choose location type:

Parks

Run

Minimum Level:

Very good

Maximum Level:

None

Radius (m):

100

Result limit:

20

Vienna POI

Air Quality Filter

Flickr Geo Image Search

Run

Choose location type:

Public swimming pool

Vienna POI

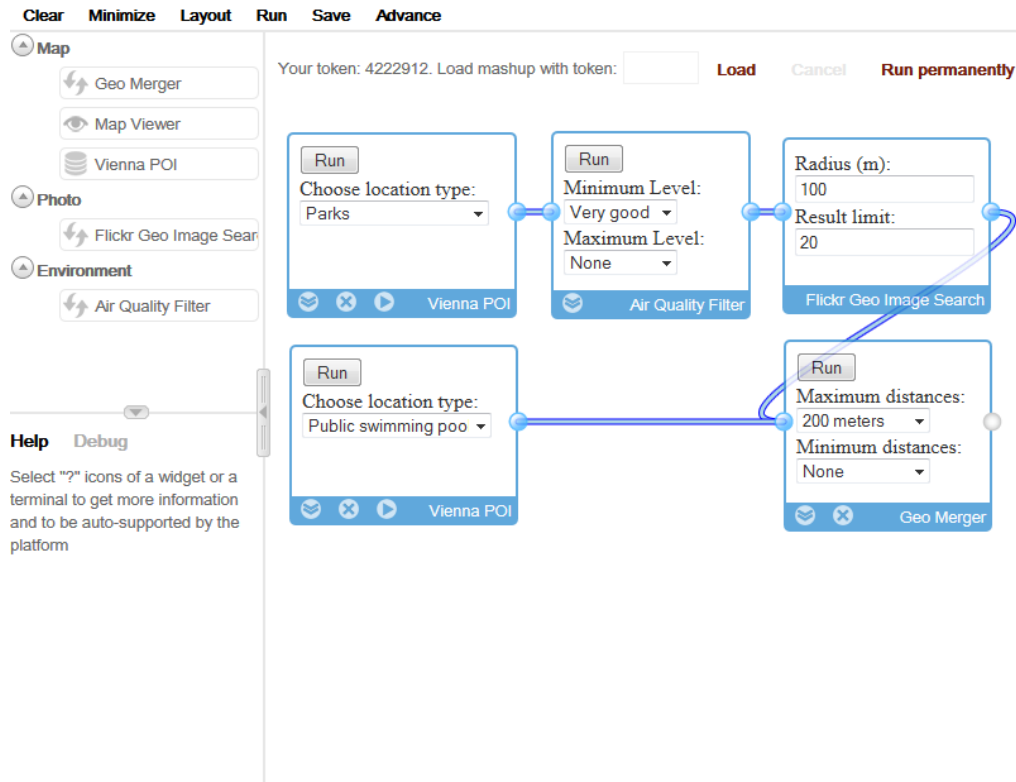
Help Debug

Select "?" icons of a widget or a terminal to get more information and to be auto-supported by the platform

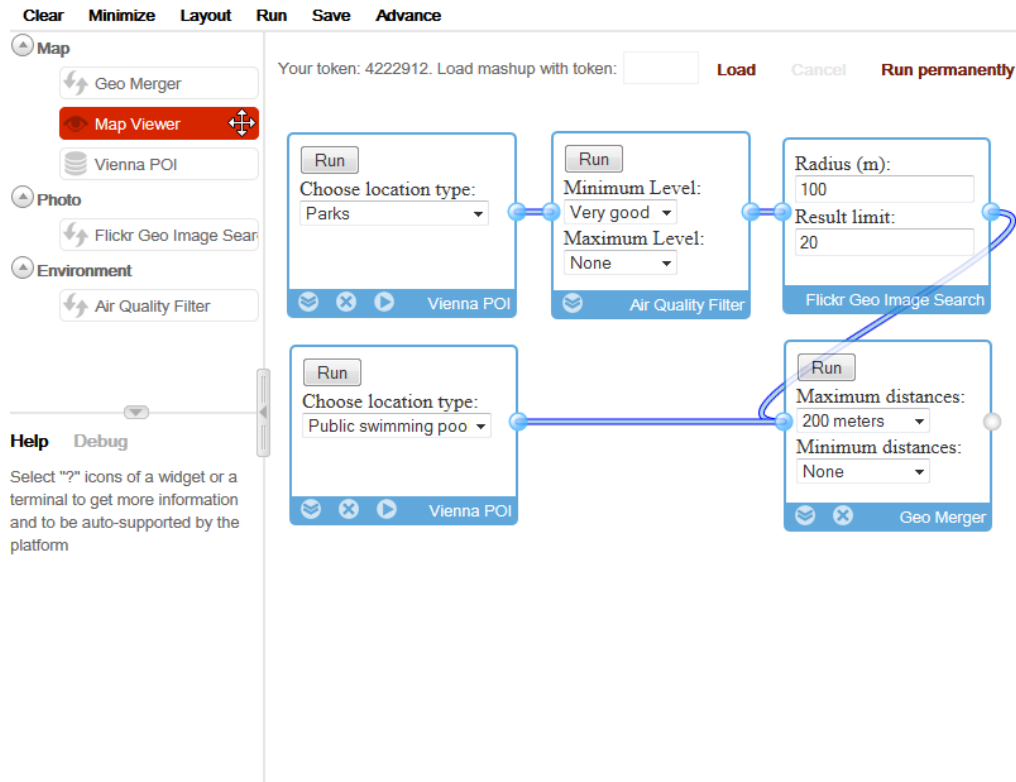
Modular approach for data integration

The screenshot displays a software interface for data integration, featuring a top menu bar with options: Clear, Minimize, Layout, Run, Save, and Advance. On the left, a sidebar lists categories: Map (Geo Merger, Map Viewer, Vienna POI), Photo (Flickr Geo Image Search), and Environment (Air Quality Filter). Below the sidebar is a 'Help' section with a 'Debug' tab and instructions: 'Select "?" icons of a widget or a terminal to get more information and to be auto-supported by the platform'. The main workspace shows a workflow of five widgets connected by lines. The first row contains three widgets: 'Vienna POI' (with 'Choose location type' set to 'Parks'), 'Air Quality Filter' (with 'Minimum Level' set to 'Very good' and 'Maximum Level' set to 'None'), and 'Flickr Geo Image Search' (with 'Radius (m)' set to '100' and 'Result limit' set to '20'). The second row contains two widgets: 'Vienna POI' (with 'Choose location type' set to 'Public swimming pool') and 'Geo Merger' (with 'Maximum distances' set to '200 meters' and 'Minimum distances' set to 'None'). A mouse cursor is pointing at the 'Geo Merger' widget. At the top of the workspace, there is a token input field with the text 'Your token: 4222912. Load mashup with token:' and buttons for 'Load', 'Cancel', and 'Run permanently'.

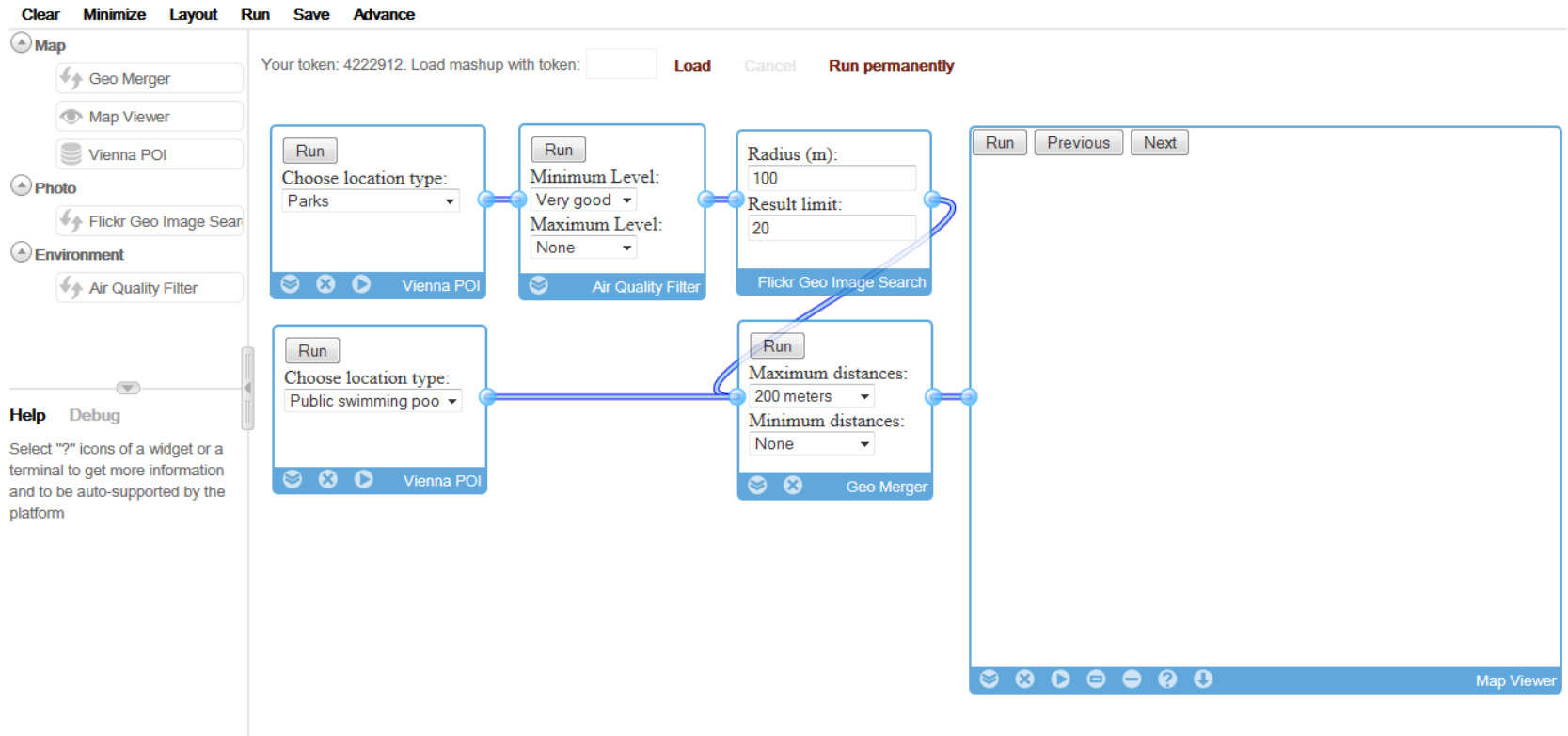
Modular approach for data integration



Modular approach for data integration



Modular approach for data integration



Modular approach for data integration

The screenshot displays the OpenStreetMap Mashup Maker interface, illustrating a modular approach for data integration. The interface is divided into several sections:

- Top Bar:** Contains buttons for 'Clear', 'Minimize', 'Layout', 'Run', 'Save', and 'Advance'. Below these, it shows 'Your token: 4222912. Load mashup with token: [input field] Load Cancel Run permanently'.
- Left Sidebar:** Lists categories of widgets: 'Map' (Geo Merger, Map Viewer, Vienna POI), 'Photo' (Flickr Geo Image Search), and 'Environment' (Air Quality Filter). Below this is a 'Help Debug' section with instructions: 'Select "?" icons of a widget or a terminal to get more information and to be auto-supported by the platform'.
- Workflow Canvas:** The central area where widgets are connected. It shows two parallel flows:
 - Top Flow:** A 'Vienna POI' widget (with 'Choose location type: Parks') is connected to an 'Air Quality Filter' widget (with 'Minimum Level: Very good' and 'Maximum Level: None'), which is then connected to a 'Flickr Geo Image Search' widget (with 'Radius (m): 100' and 'Result limit: 20').
 - Bottom Flow:** A 'Vienna POI' widget (with 'Choose location type: Public swimming poo') is connected to a 'Geo Merger' widget (with 'Maximum distances: 200 meters' and 'Minimum distances: None').
- Map Viewer:** A large empty map area on the right, with 'Run', 'Previous', and 'Next' buttons at the top. At the bottom, there is a toolbar with icons for map navigation, and a mouse cursor is pointing at the play button icon.

Modular approach for data integration

The screenshot displays a modular data integration interface with a top menu bar (Clear, Minimize, Layout, Run, Save, Advance) and a left sidebar with categories: Map (Geo Merger, Map Viewer, Vienna POI), Photo (Flickr Geo Image Search), and Environment (Air Quality Filter). A 'Help' section at the bottom left explains the '?' icon. The main workspace shows a workflow with six modules: two 'Vienna POI' modules, one 'Air Quality Filter', one 'Flickr Geo Image Search', and one 'Geo Merger'. The workflow is as follows: 1. 'Vienna POI' (Choose location type: Parks) connects to 2. 'Air Quality Filter' (Minimum Level: Very good, Maximum Level: None), which connects to 3. 'Flickr Geo Image Search' (Radius: 100m, Result limit: 20). This sequence connects to 4. 'Vienna POI' (Choose location type: Public swimming pool), which connects to 5. 'Geo Merger' (Maximum distances: 200m, Minimum distances: None). The final output is shown in a 'Map Viewer' window on the right, displaying a map of Vienna with a green pin and a data panel. The data panel shows: Found: 33, Current: 3; Map/Satellite view; Left/Top/Bottom/Right/Full Map controls; Metadata including @type, @id, location, @type, lat, long, address, and airQuality; and two image thumbnails of a park.

Clear Minimize Layout Run Save Advance

Map

Geo Merger

Map Viewer

Vienna POI

Photo

Flickr Geo Image Search

Environment

Air Quality Filter

Help Debug

Select "?" icons of a widget or a terminal to get more information and to be auto-supported by the platform

Your token: 4222912. Load mashup with token: Load Cancel Run permanently

Run

Choose location type: Parks

Run

Minimum Level: Very good

Maximum Level: None

Run

Radius (m): 100

Result limit: 20

Run

Choose location type: Public swimming pool

Run

Maximum distances: 200 meters

Minimum distances: None

Run

Map Viewer

Map Satellite

Found: 33. Current: 3

Left Top Bottom Right Full Map

@type: "http://ogd.ifs.tuwien.ac.at/vienna/PARKANLAGEOGD"

@id: "http://ogd.ifs.tuwien.ac.at/vienna/PARKANLAGEOGD.74802"

location: Object

@type: "http://www.w3.org/2003/01/geo/wgs84_pos#Point"

lat: "48.179088632252956"

long: "16.317551227631565"

address: "Schönbrunner Schloßstraße 47, 1130 Wien, Austria"

airQuality: 0.8257720583919212

Mashup-based data integration

- Mashups: innovative paradigm that “*combines data from multiple sources into an **integrated** and **single graphical interface***” [1]
 - are quick, flexible, and cost-effective
 - turn users from content consumers to content providers

Challenges of mashup-based data integration research

1. Collaborative work among **end-user**, **data publisher** and **developer** communities is not considered
2. Integrating data that is **distributed** in different devices and not available on the web is not considered
3. **Semantic mashup** – “*A data mashup using RDF(S) as data model*” [2] – is still in its early stage

[2] Endres-Niggemeyer, Brigitte. *Semantic mashups: Intelligent Reuse of Web Resources* (2013)

Research Question

*How can **non-expert** users be enabled to explore and integrate **heterogeneous** data sources?*

RQ1: *How is it possible to support non-expert users in addressing data **heterogeneity**?*

RQ2: *How can non-expert users be enabled to **collaboratively** integrate data?*

RQ3: *How is it possible to **automate** the data exploration and integration process?*

2. Linked Widgets Framework

Linked Widgets

- extends W3C standard web widgets with a semantic model
- lift data sources to a semantic level to facilitate data processing
- can be created by **independent** developers, hosted on different servers

Processing function

Semantic model

User Interface

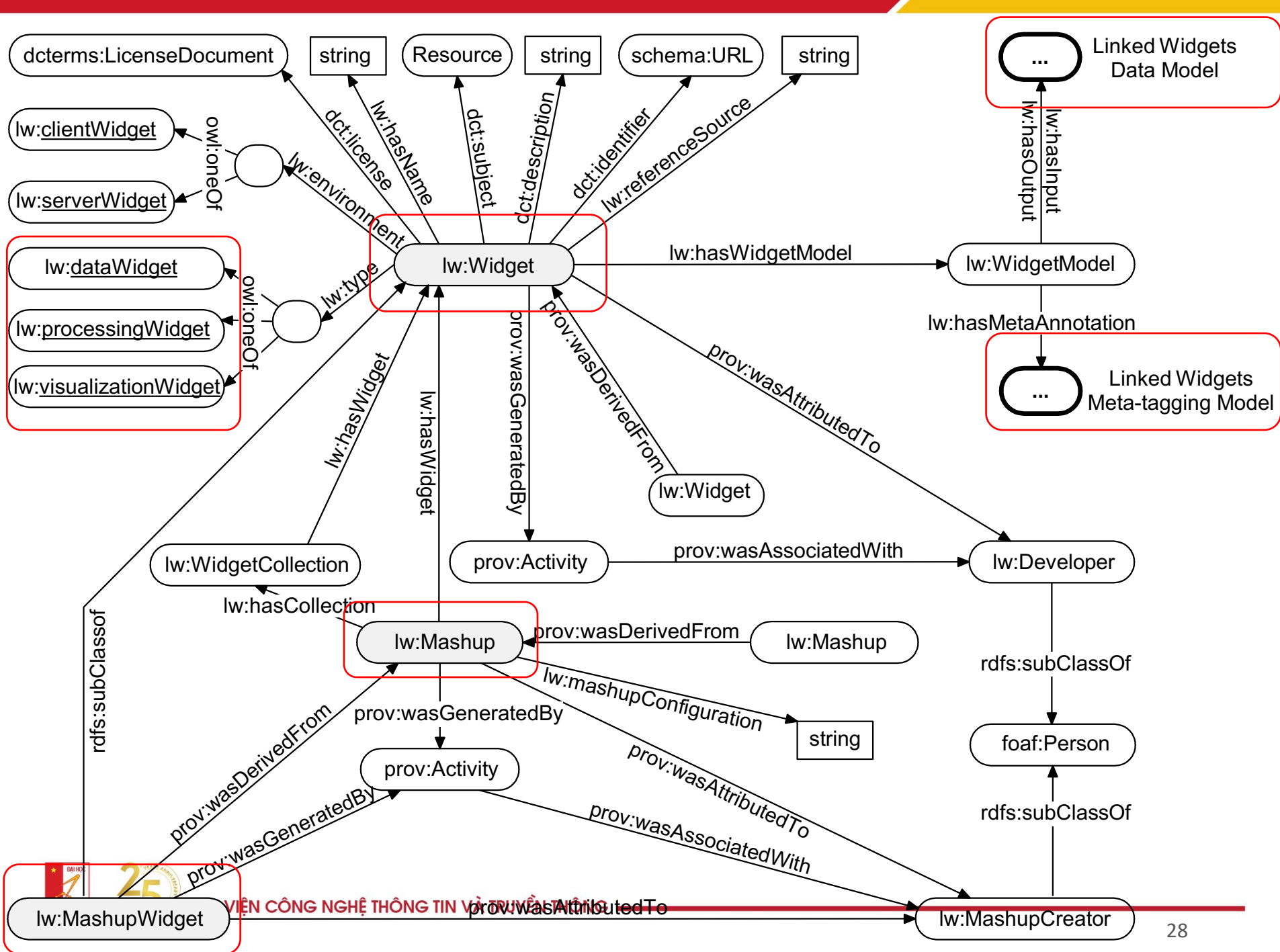
Core Interface

The diagram shows a widget titled "Point Of Interest Search". It has a "Semantic model" section with "POI type:" (set to "park") and "Radius (km)" (with a dropdown menu showing "MotorcycleParking", "NationalPark", "Park", "ParkBench", and "Parking"). Below this is a "User Interface" section with a search bar. At the bottom is a "Core Interface" section with a blue bar containing icons for search, cancel, play, and stop, and the text "Point Of Interest Search".

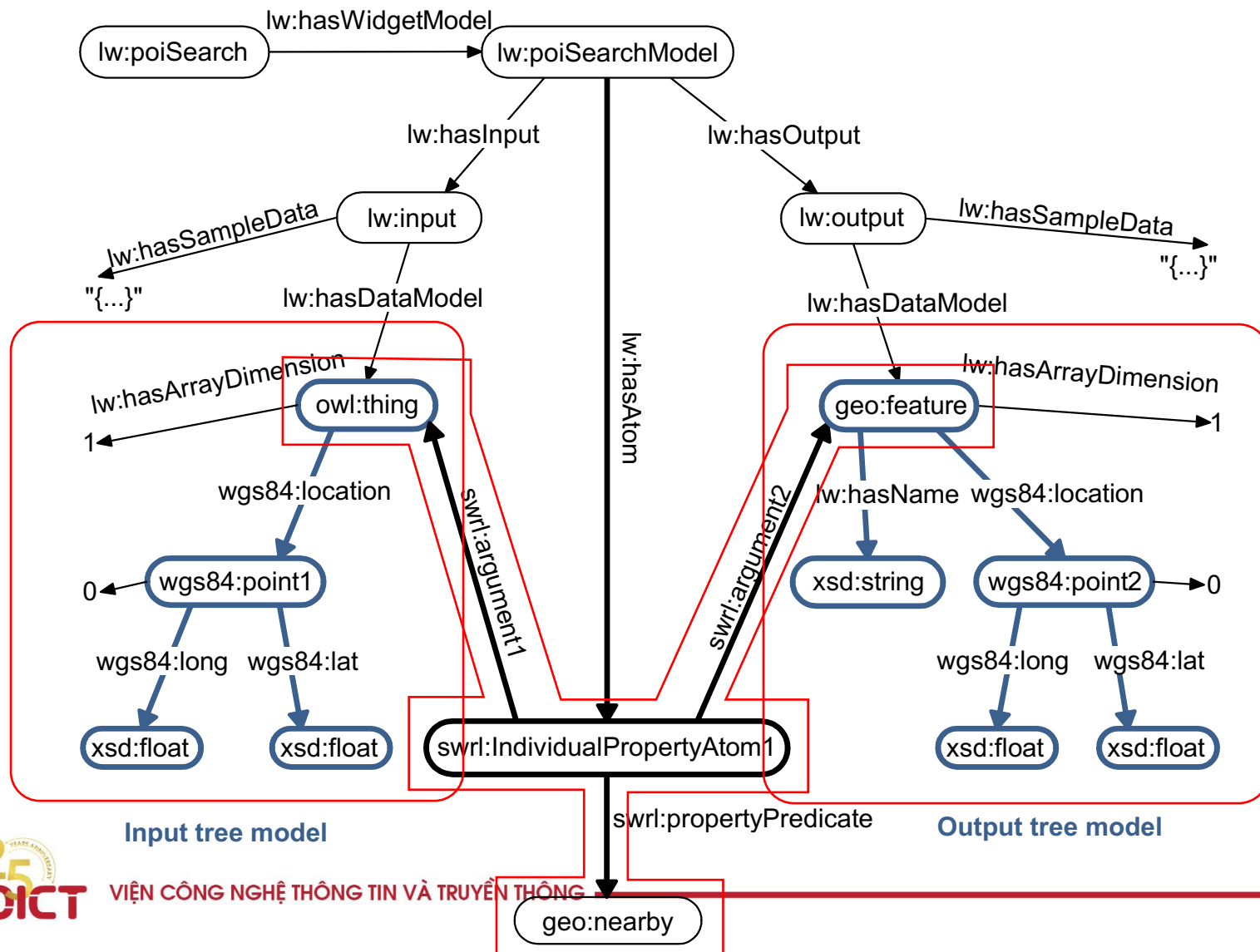
Client Linked Widgets
run in a browser

The diagram shows a widget titled "Weather Observation". It has a "Semantic model" section with "Indicator" (with radio buttons for "Temperature", "Pressure", and "Wind") and "Timer:" (set to "5 seconds" with a dropdown arrow). Below this is a "User Interface" section with a green bar containing icons for search, cancel, play, and stop, and the text "Weather Observation".

Server Linked Widgets run in a
web server, PC, mobile phones,
sensor, or embedded device



Example: Data model of the POI search widget



M1: Semantic Widget Discovery

Clear

Show previous results

Exact Search

Tolerant Search

☒ Short Name ☐ Full URI

To define widget's model, please drag & drop Data Model Relation and if necessary, use keywords

+ Terminal

Data Model

☒ Input

☒ Output

Type: of

+ with attributes & relations with others

☒ location

Data Model

Type: of

+ with attributes & relations with others

☒ lat

☒ long

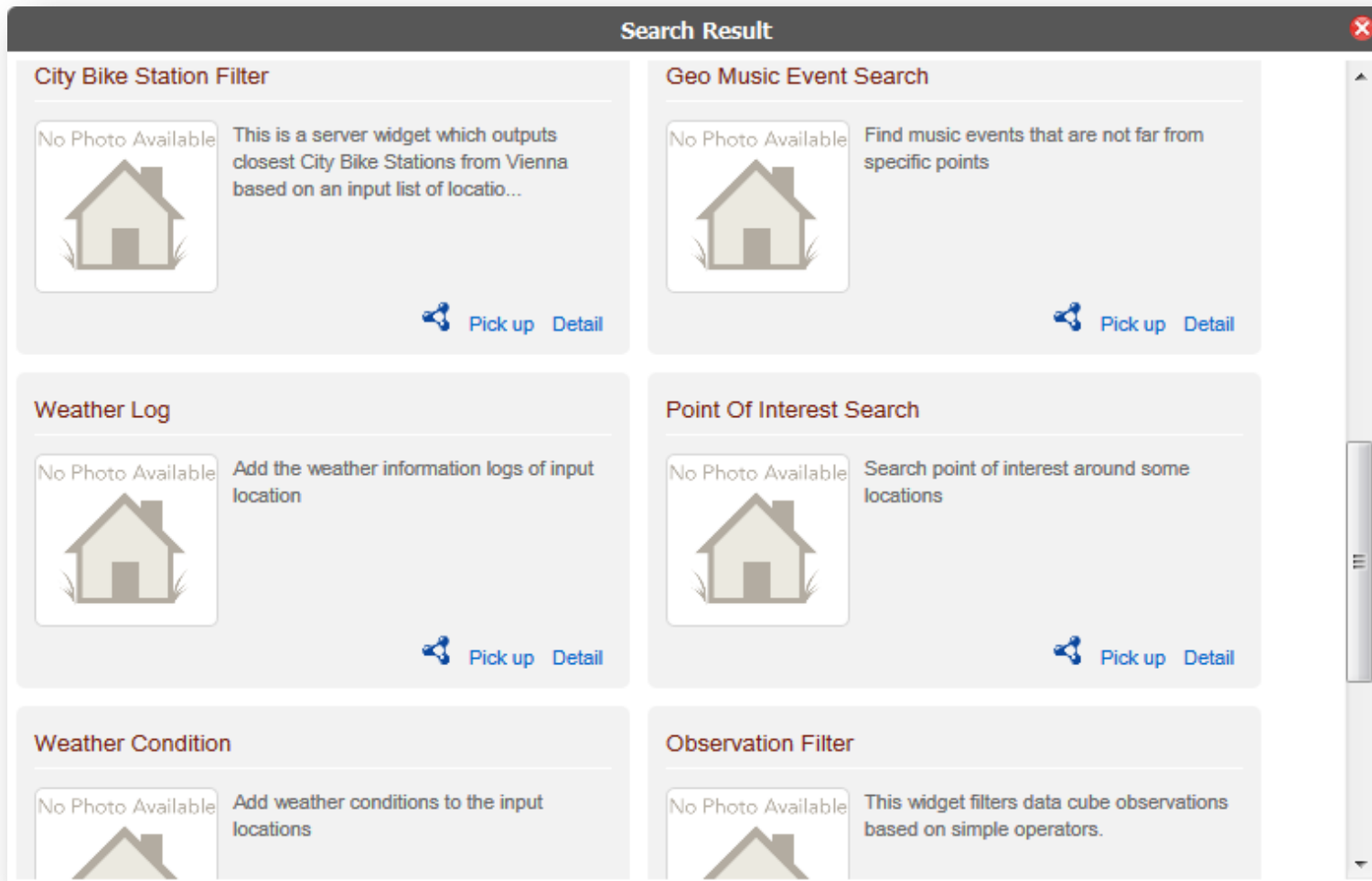
Point

Data Model

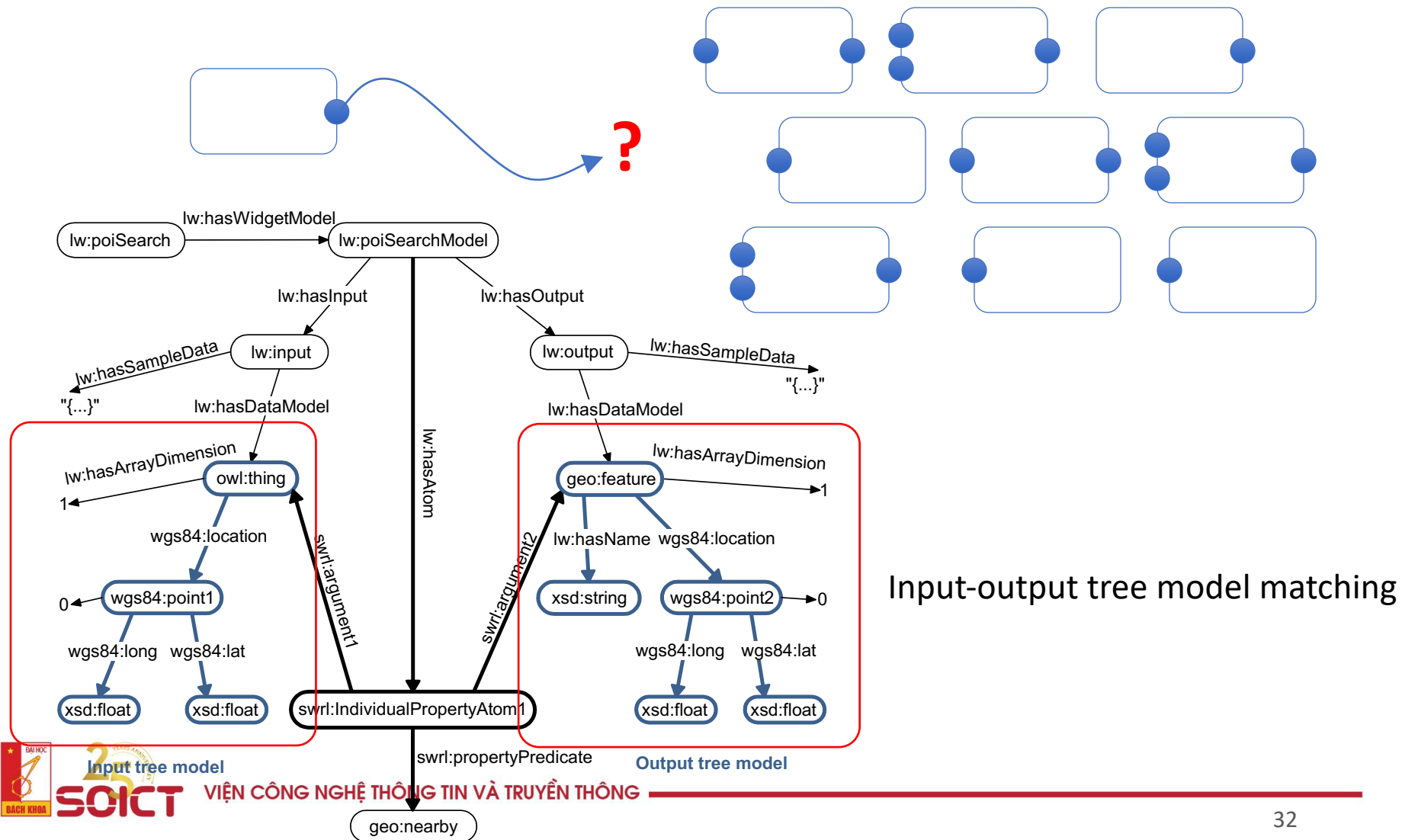
Ontology Alignment API:

<http://alignapi.gforge.inria.fr/>

M1: Semantic Widget Discovery



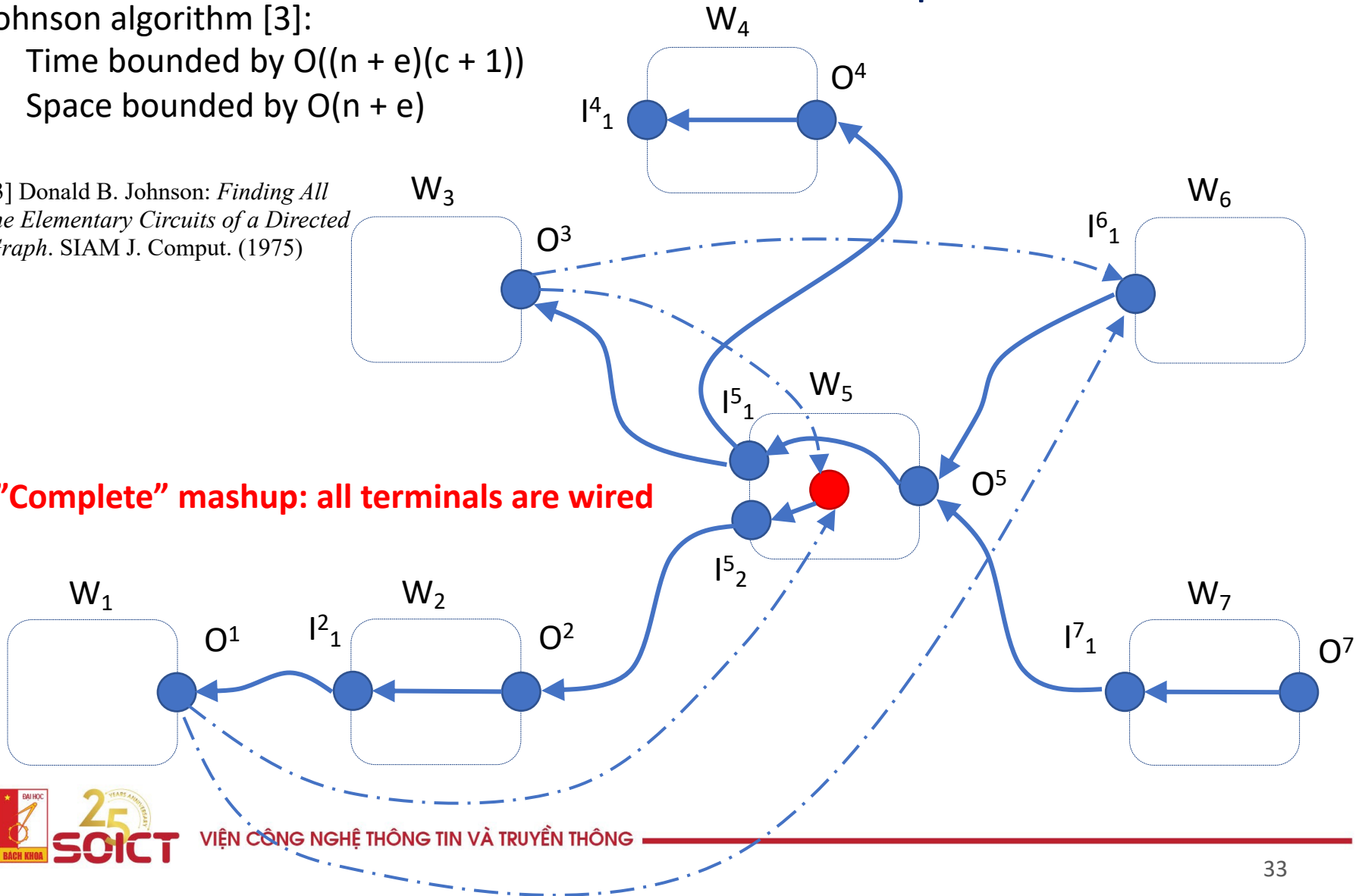
M2: Terminal matching



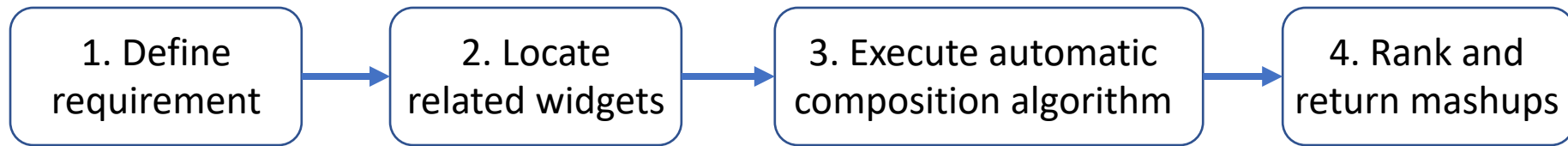
Reduce “*automatic composition*” to “*list all cycles*” algorithm

- Time bounded by $O((n + e)(c + 1))$
- Space bounded by $O(n + e)$

"Complete" mashup: all terminals are wired



M4: Interactive tag-based composer



Enter text + tie words to
DBpedia (Wikipedia) resources

Restaurant nearby Bank



<http://dbpedia.org/resource/Restaurant>



<http://www.geonames.org/ontology#nearby>



<http://dbpedia.org/resource/Bank>

A bank is a financial intermediary that creates credit by lending money to a borrower, thereby creating a corresponding deposit on the bank's balance sheet etc.

Benefits of the semantic tagging techniques

- No ambiguity
- Multiple language

Define mashup requirement

☒ Free Style ☐ Syntax Style **Compose Mashup**

Restaurant nearby Bank

☒ Concept ☐ Relation ☐ Who ☐ Where ☐ When ☐ What

Bank (<https://en.wikipedia.org/wiki/Bank>)
A bank is a financial institution that creates credit by lending money...

Bank of America (https://en.wikipedia.org/wiki/Bank_of_America)
Bank of America (abbreviated as BofA) is an American multinational bank...

Bankruptcy (<https://en.wikipedia.org/wiki/Bankruptcy>)
Bankruptcy is a legal status of a person or other entity that cannot r...

Bank and Monument stations (https://en.wikipedia.org/wiki/Bank_and_Monument_stations)
Bank and Monument are interlinked London Underground and Docklands Lig...

Banksia (<https://en.wikipedia.org/wiki/Banksia>)
Banksia is a genus of around 170 species in the plant family Proteacea...

1. **Concept tagging**
(DBpedia concepts)

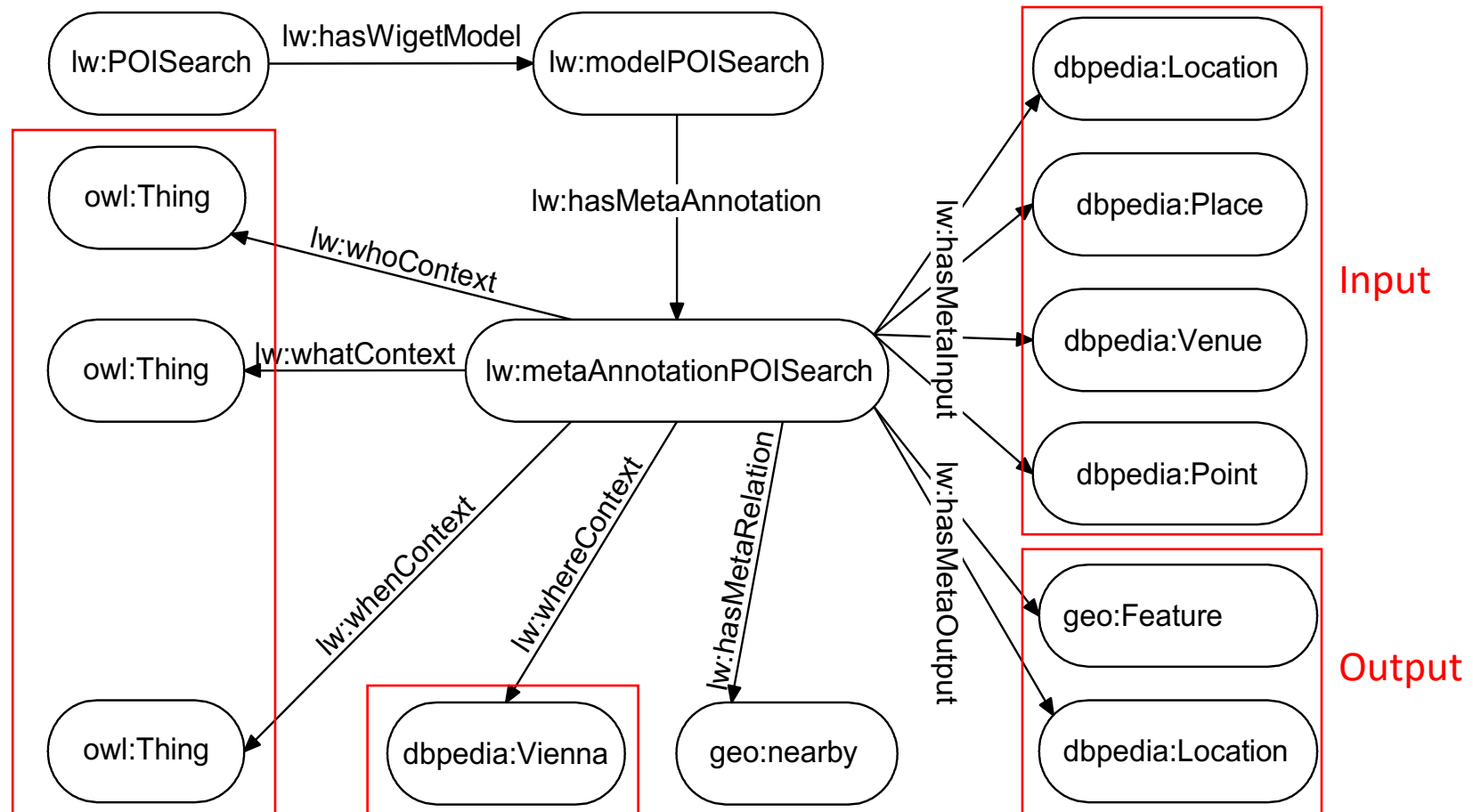
2. **Relation tagging**
(LOD properties)

3. **Entity tagging**
DBpedia entities

Example queries:

1. Point of interest
2. Vienna Castles
3. Playground with

Meta-tagging model



lw <http://linkedwidgets.org/ontologies/>

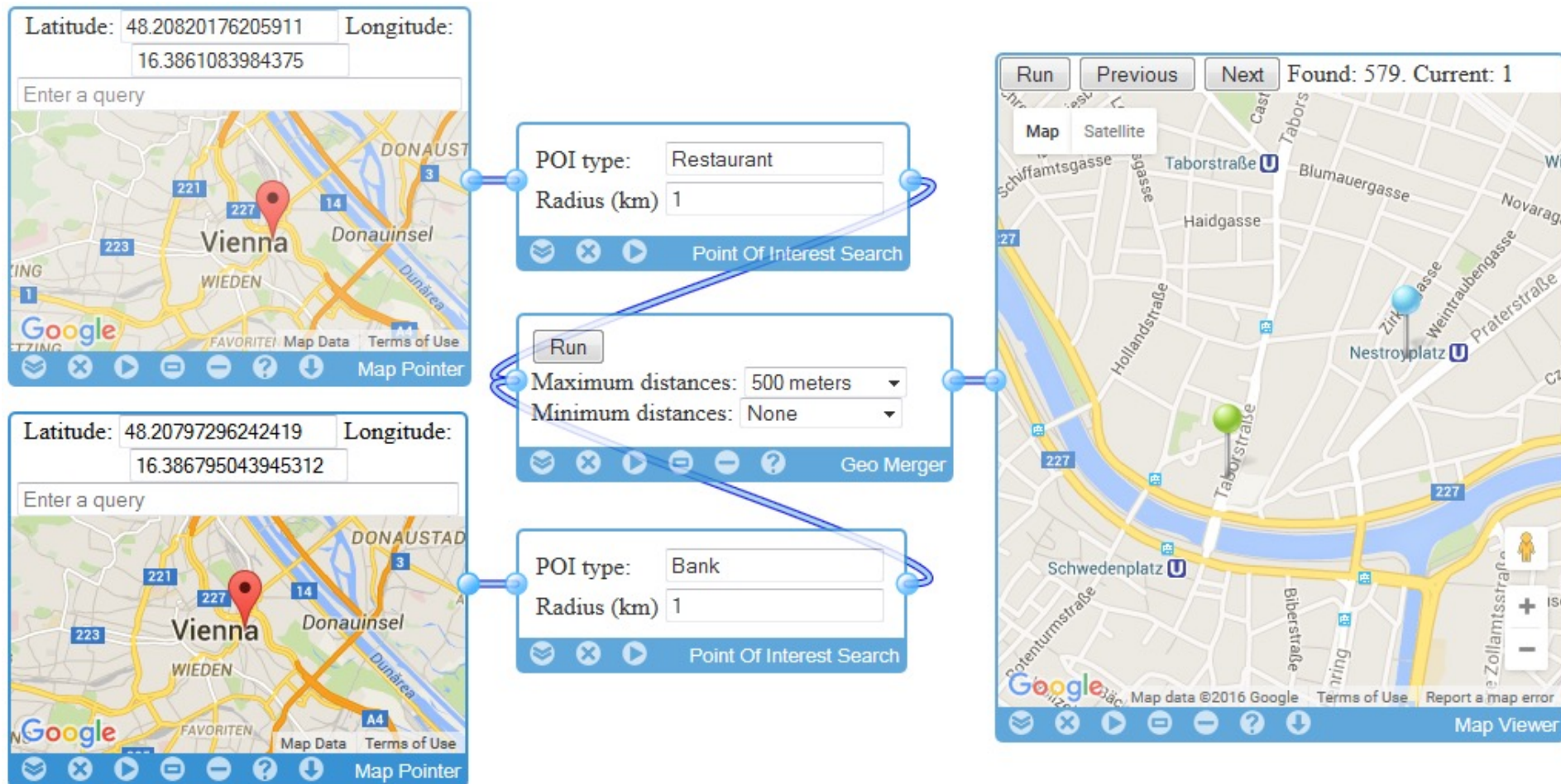
owl <http://www.w3.org/2002/07/owl#>

dbpedia <http://dbpedia.org/resource/>

geo <http://www.geonames.org/ontology#>

Example

4 widgets: Map pointer, Geo merger, POI search, Map viewer

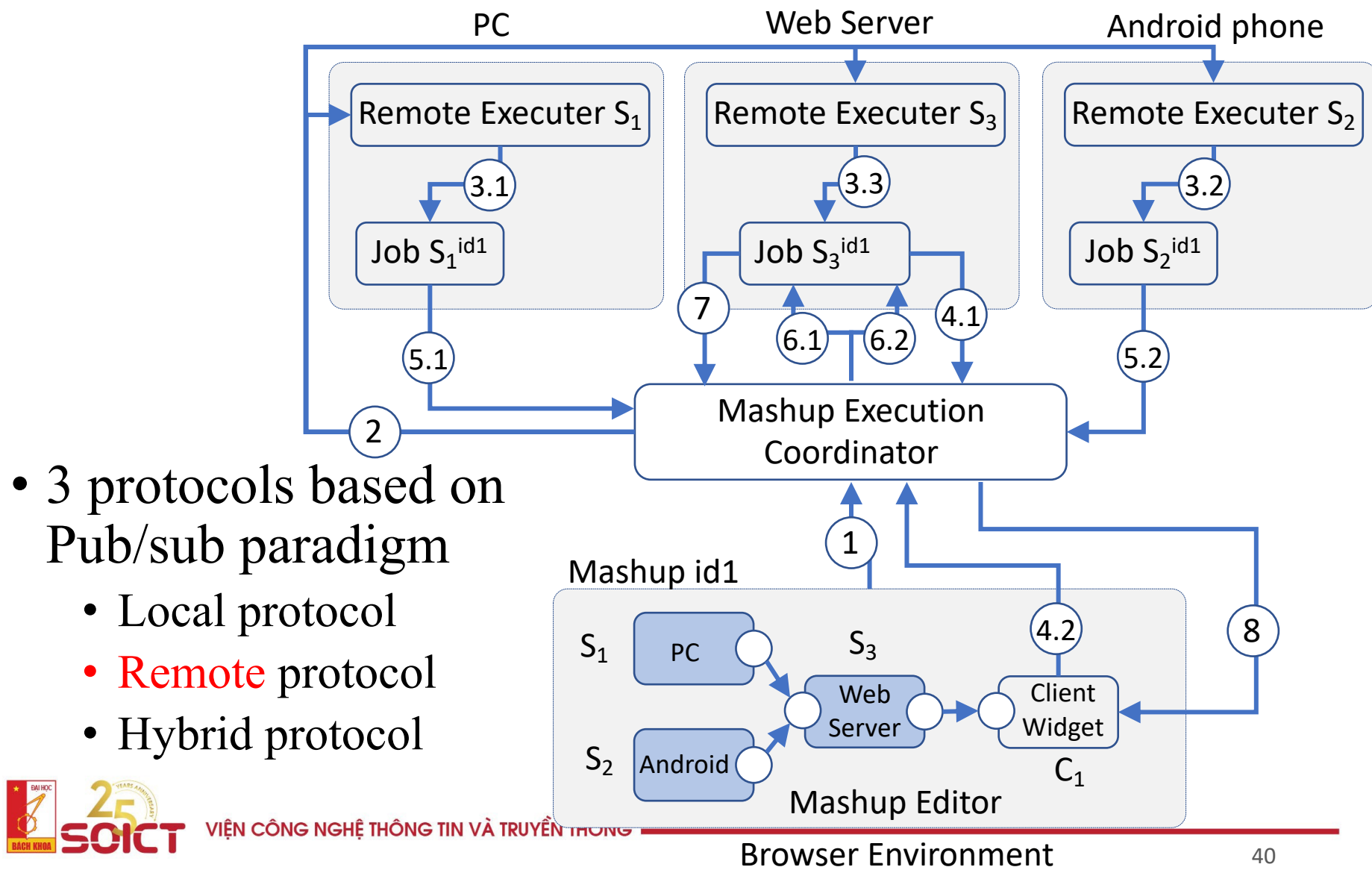


3. Mashup Models

3. Mashup models

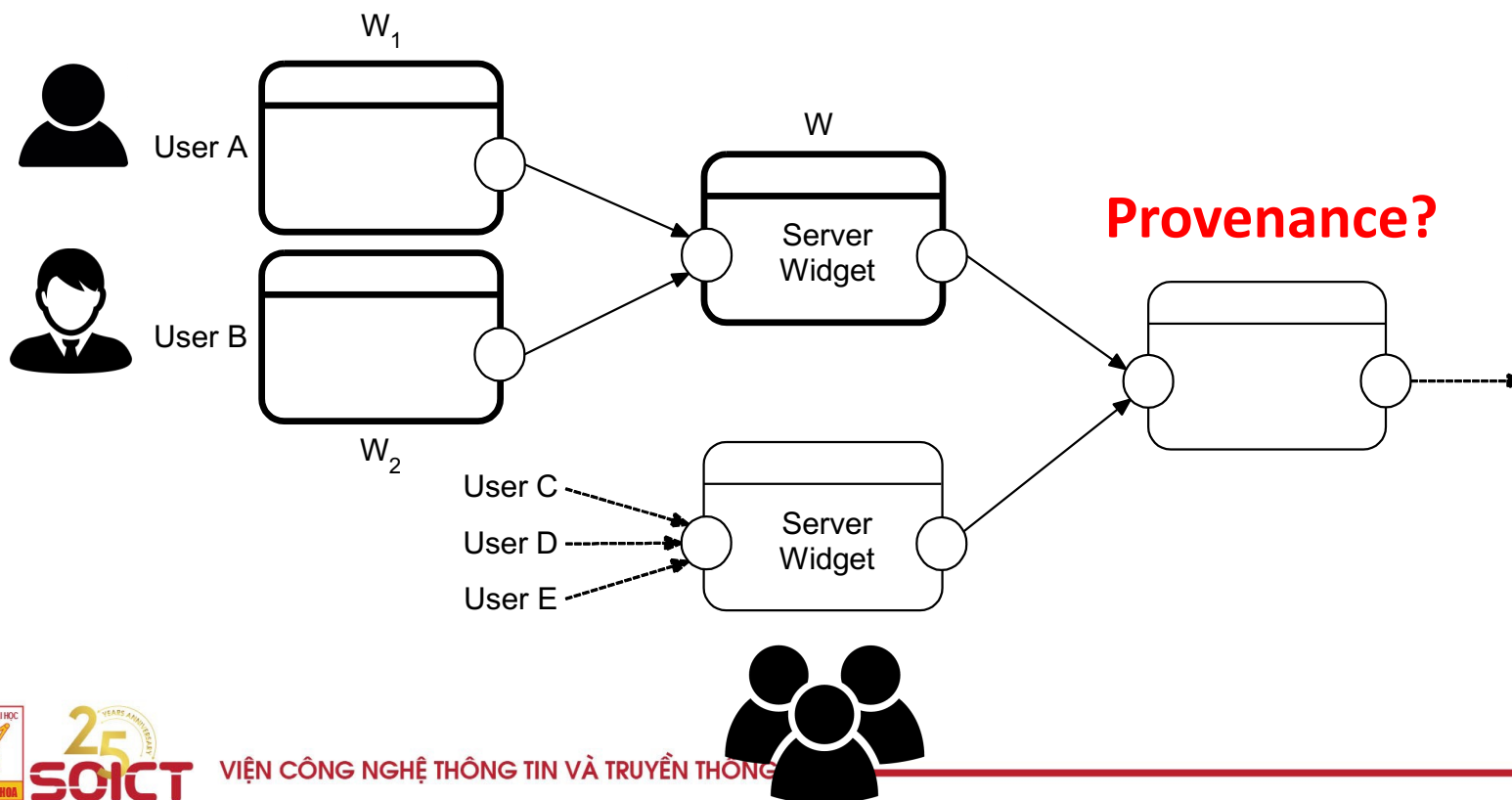
- Model 1: collaborative mashups
- Model 2: persistent mashups
- Model 3: streaming mashups
- Model 4: distributed mashups

Linked Widget communication protocols



Model 1 – Collaborative mashups

- Mashups are created and/or operated by multiple users at the same time



Processing Unit

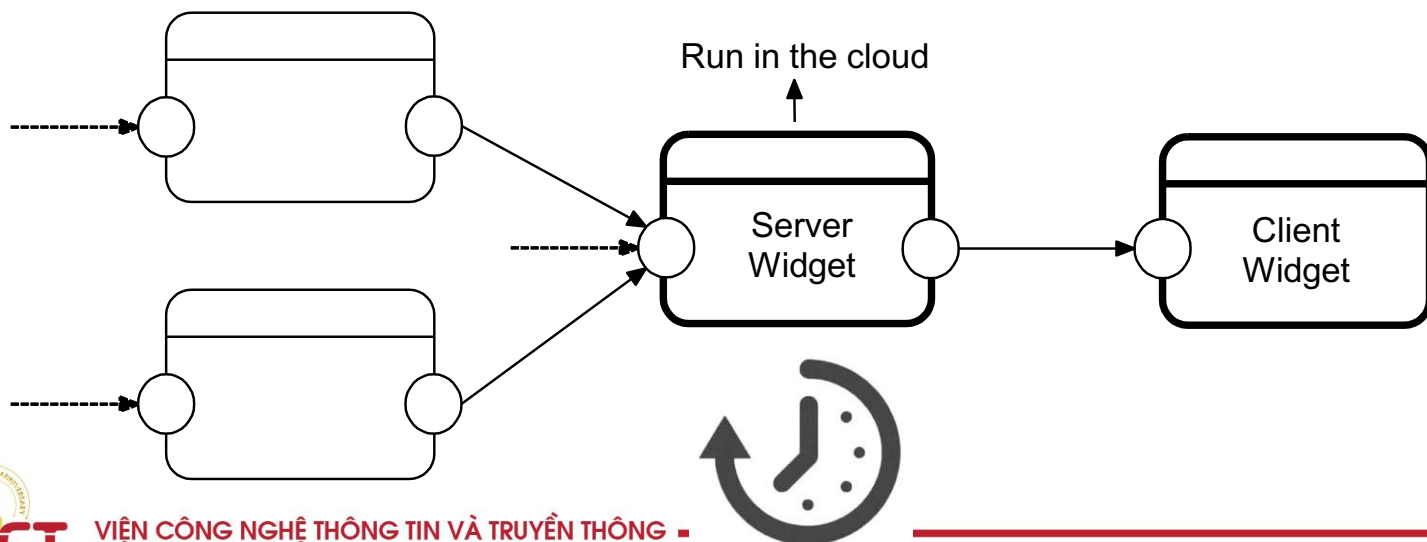


xsd:string ← signature



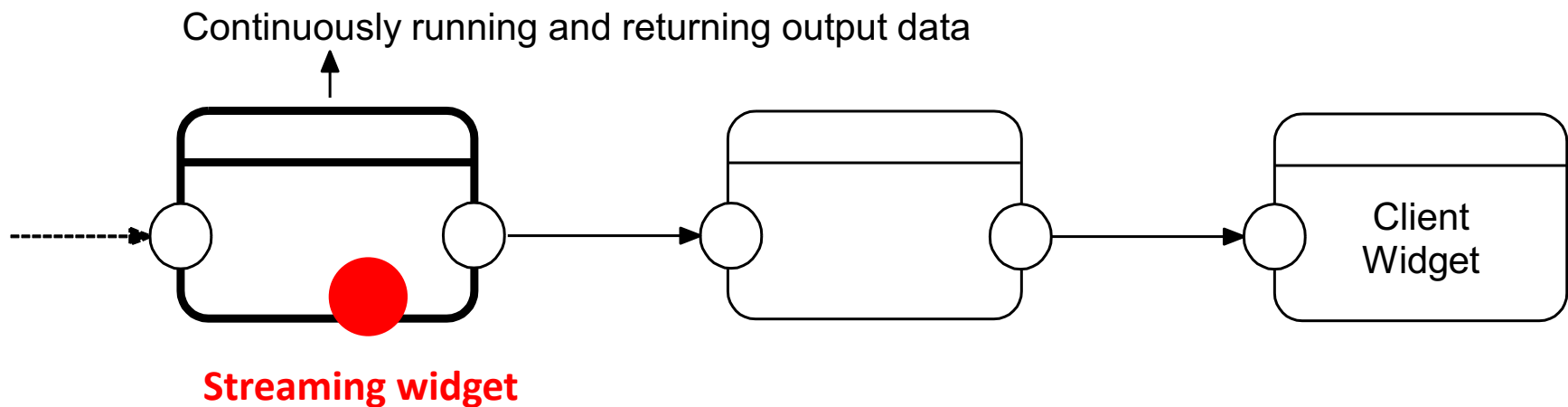
Model 2 – Persistent mashups

- run continuously in the background
- are useful for **time-consuming** data integration tasks



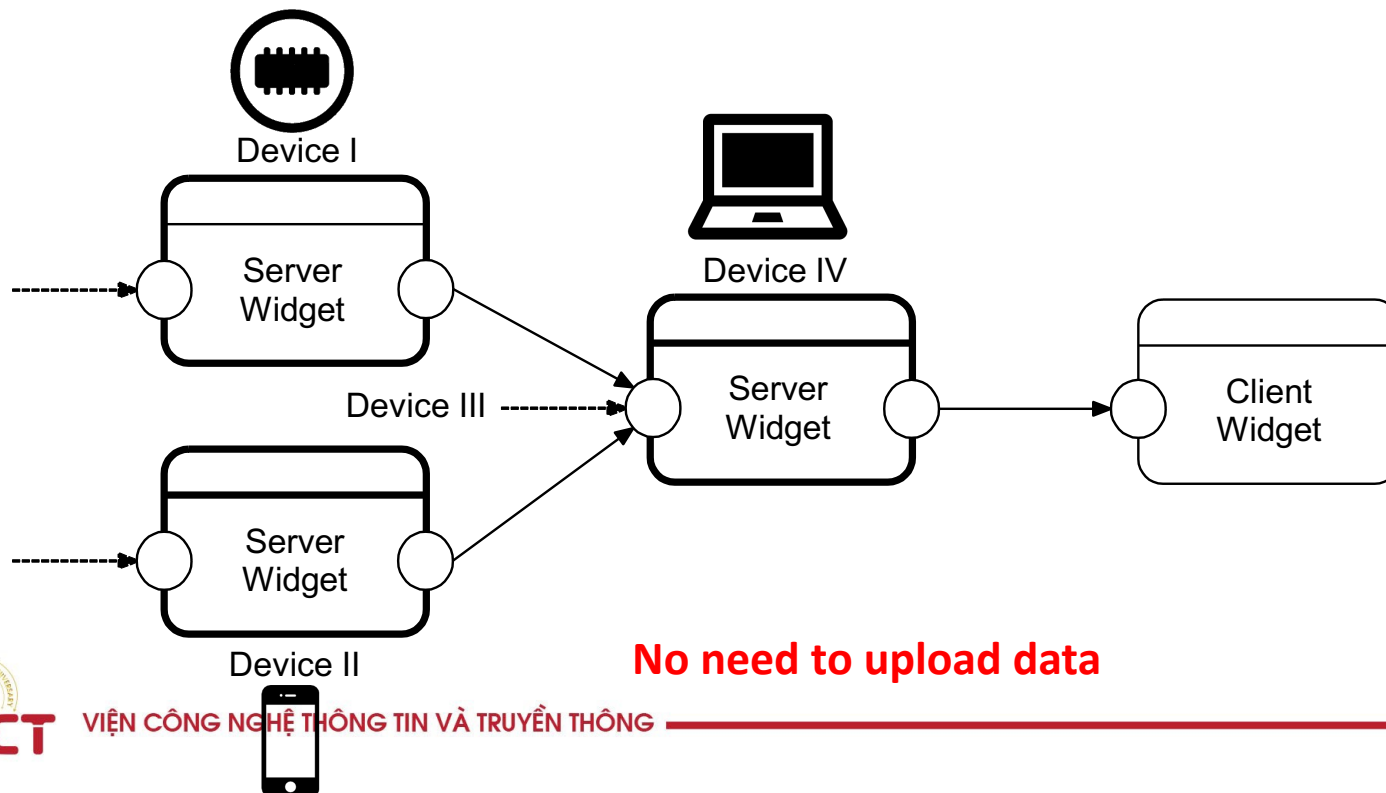
Model 3 – Streaming mashups

- Data is continuously flowing from a widget to others
- Streaming mashups are useful for data monitoring use cases.



Model 4 – Distributed mashups

- Widgets are hosted by distributed nodes and devices
- Useful for integrating sensor data and data from embedded devices



Distributed & collaborative mashup example: Combine and visualize sales data for a series of retail points of sale (POS)

POS Spreadsheet (pos, location, country, city)

Link to the Google sheet

```
https://docs.google.com/spreadsheets/d/1vcB8YooN1G7zaYebvI25k0hpULk0ossEUDkqJ3waIAk/edit#gid=0
```

Google Sheet

Merge multiple W3C cube datasets

Cube Merger

Filter

1. POS --
2. Date --
3. Category Fruit
4. Location --
5. Country --
6. City --
7. Shop --

Function: ☒ Sum ☐ Average

1. ☐ POS
2. ☐ Date
 - ☐ Monthly
 - ☐ Quarterly
 - ☐ Yearly
3. ☐ Category
4. ☐ Location
5. ☐ Country
6. ☒ City

Aggregation

Please remember to run your spreadsheet server widget first!

POS 1

Please remember to run your spreadsheet server widget first!

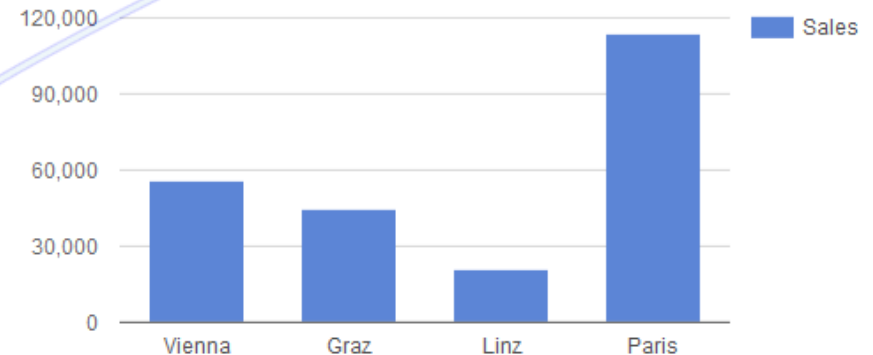
POS 2

Please remember to run your spreadsheet server widget first!

POS 3

Run

Type of chart Column chart



Sales Spreadsheets (pos, date, category, sale)

4. Conclusions and Future Work

Conclusions

- Development of concepts combining semantic web and mashups
- Semantic model for widgets
- Facilitating collaborative work among **end-user**, **data publisher** and **developer** communities
- **Introducing a new model** of *semantic*, *distributed*, and *collaborative* mashups
- Running prototype: <http://linkedwidgets.org>